

University of California Riverside
Electrical and Computer Engineering Department

Electromagnetics I

Lecture 4: Maxwell's Equations in Differential Forms

Chapter 2

04/09/2026

Dr. Alex Khitun

Email: akhitun@engr.ucr.edu

Office Hours: Thursday, 8:00 AM - 9:00 AM, MSE 016

Outline

- Scalar vs. Vector Fields
- The Del operator
- Gradient of a Scalar Field $\vec{\nabla}V$
- Divergence of a Vector Field $\vec{\nabla} \cdot \vec{V}$
- Curl of a Vector Field $\vec{\nabla} \times \vec{V}$
- Maxwell's equations in differential forms
- Divergence theorem
- Stokes' theorem
- Electrostatic potential

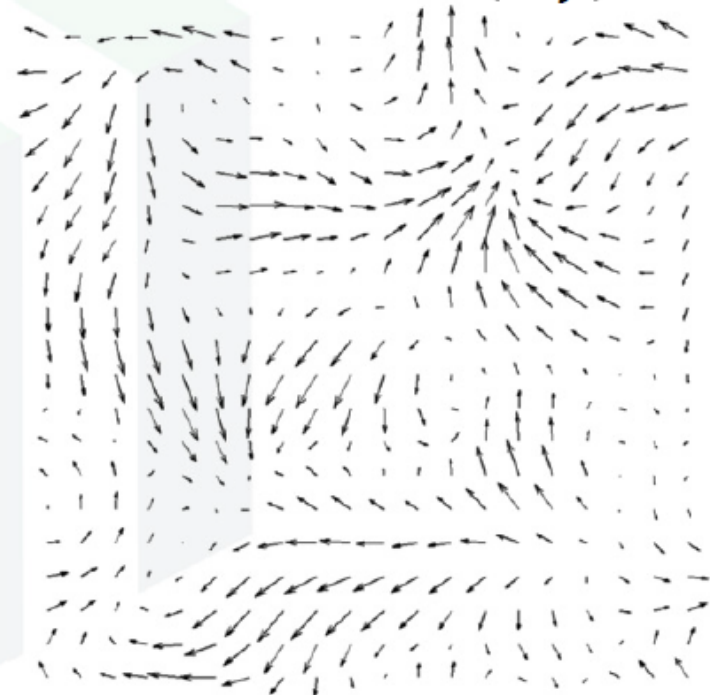
Scalar Field Vs. Vector Field (2D)

Scalar Field $f(x, y)$



- Magnitude only

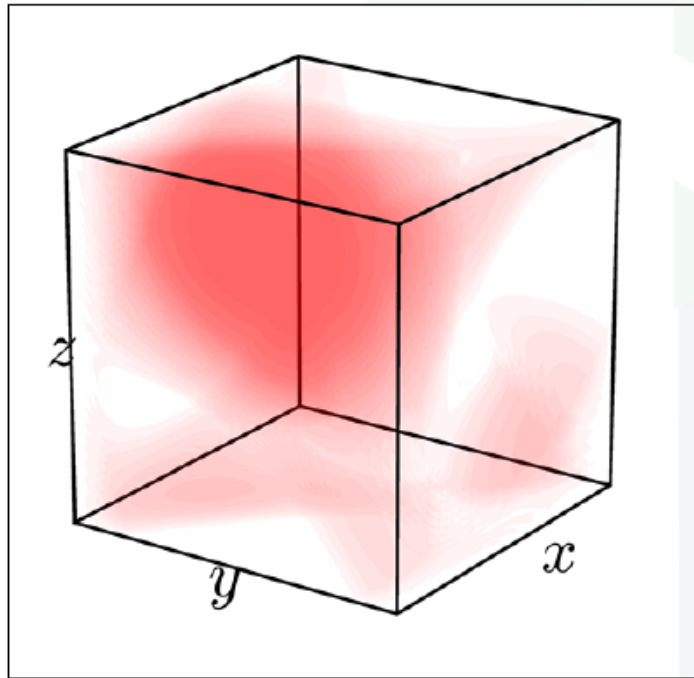
Vector Field $\vec{F}(x, y)$



- Magnitude
- Direction

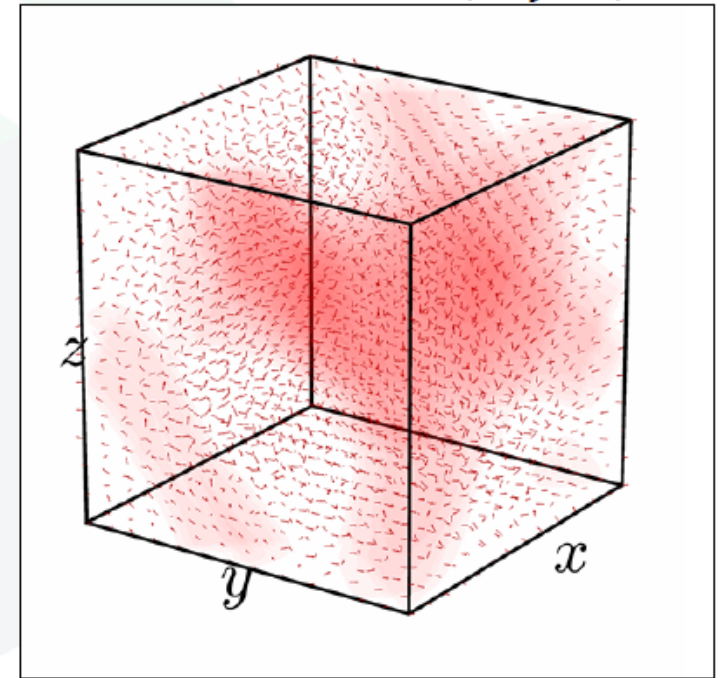
Scalar Field Vs. Vector Field (3D)

Scalar Field $f(x, y, z)$



- Magnitude only

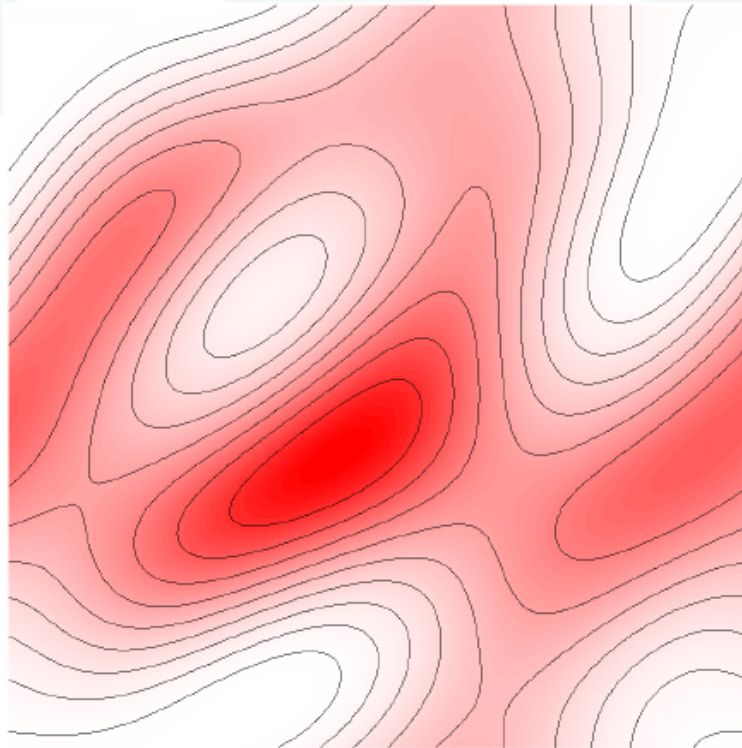
Vector Field $\vec{F}(x, y, z)$



- Magnitude
- Direction

Isocontour Lines

Isocontour lines trace the paths of equal value. Closely spaced isocontours convey that the function is varying rapidly.



The Del Operator ∇

The del operator ∇ is the vector differential operator. It is sort of a 3D derivative.

Coordinate System	Del Operator
Cartesian	$\nabla = \frac{\partial}{\partial x} \hat{a}_x + \frac{\partial}{\partial y} \hat{a}_y + \frac{\partial}{\partial z} \hat{a}_z$
Cylindrical	$\nabla = \frac{\partial}{\partial \rho} \hat{a}_\rho + \frac{1}{\rho} \frac{\partial}{\partial \phi} \hat{a}_\phi + \frac{\partial}{\partial z} \hat{a}_z$
Spherical	$\nabla = \frac{\partial}{\partial r} \hat{a}_r + \frac{1}{r} \frac{\partial}{\partial \theta} \hat{a}_\theta + \frac{1}{r \sin \theta} \frac{\partial}{\partial \phi} \hat{a}_\phi$

Even though ∇ is a vector, it is never written as $\vec{\nabla}$.

Derived from the Cartesian equation using coordinate transformation.

Summary of Vector Operations

Operation		Input	Output
Vector Addition & Subtraction	$\vec{U} + \vec{V}$	Vectors	Vector
Dot Product	$\vec{U} \bullet \vec{V}$	Vectors	Scalar
Cross Product	$\vec{U} \times \vec{V}$	Vectors	Vector
Gradient	∇f	Scalar Function	Vector Function
Divergence	$\nabla \bullet \vec{U}$	Vector Function	Scalar Function
Curl	$\nabla \times \vec{U}$	Vector Function	Vector Function
Scalar Laplacian	$\nabla^2 \mathcal{V}$	Scalar Function	Scalar Function
Vector Laplacian	$\nabla^2 \vec{U}$	Vector Function	Vector Function

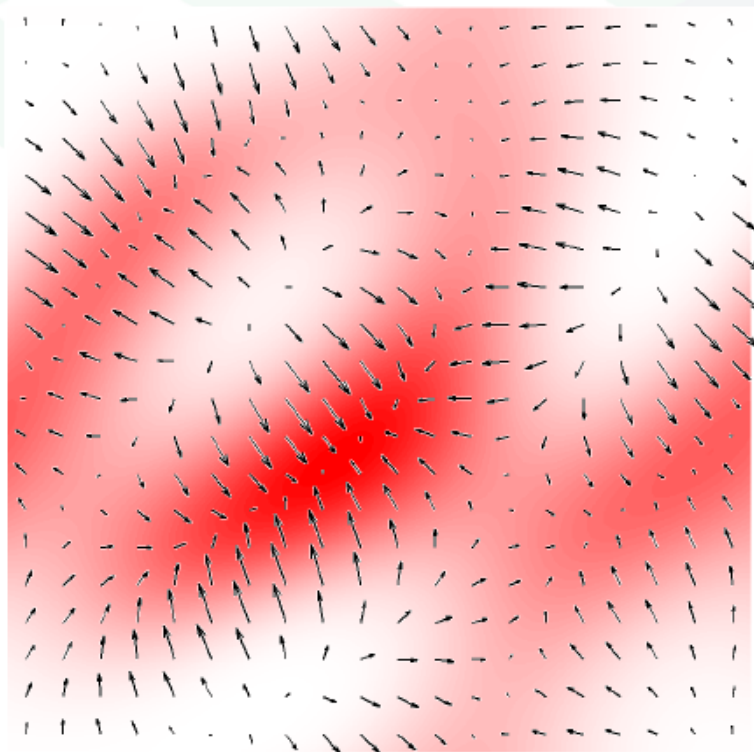
Gradient of a Scalar Field (1 of 3)

Start with a scalar field $f(x, y)$.



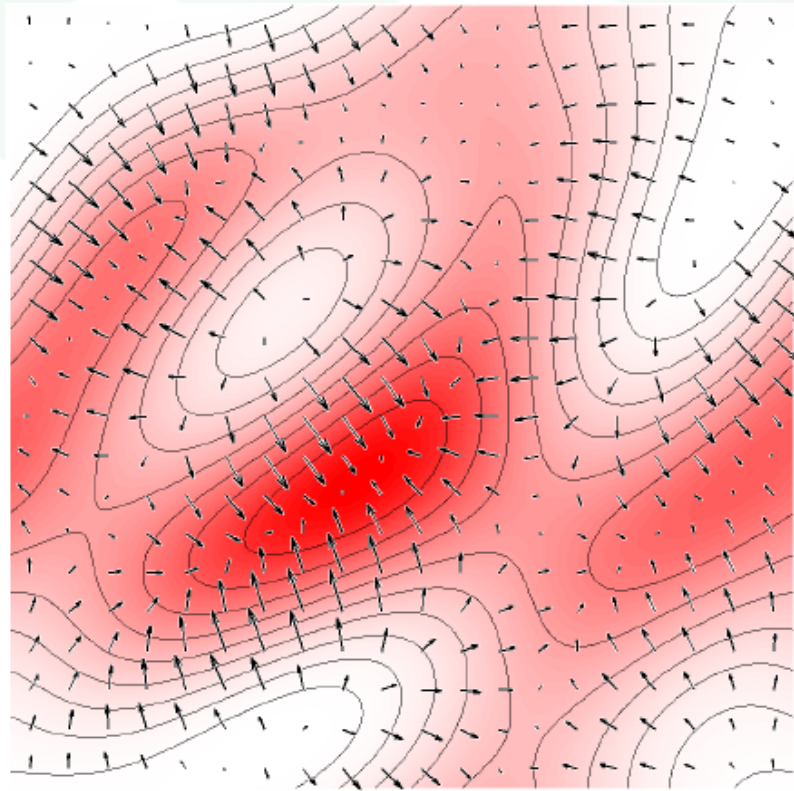
Gradient of a Scalar Field (2 of 3)

Plot the gradient $\nabla f(x, y)$ on top of $f(x, y)$. The background color is the original scalar field $f(x, y)$.



Gradient of a Scalar Field (3 of 3)

The gradient will always be perpendicular to the isocontour lines.



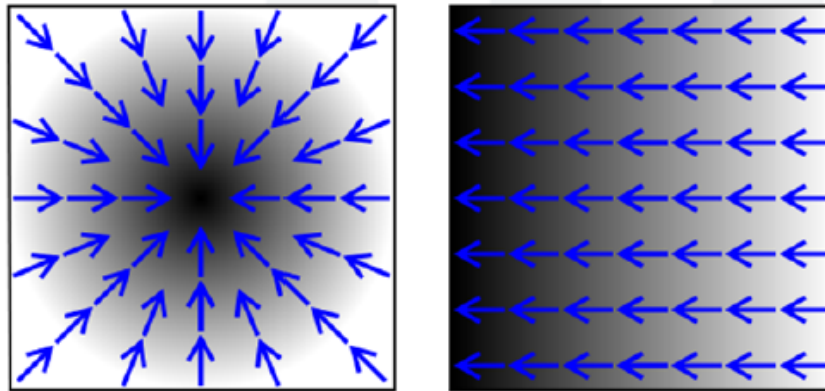
The Gradient ∇V

The gradient calculates how rapidly, and in what direction, a scalar function is increasing.

Coordinate System	Gradient Operator
Cartesian	$\nabla V = \frac{\partial V}{\partial x} \hat{a}_x + \frac{\partial V}{\partial y} \hat{a}_y + \frac{\partial V}{\partial z} \hat{a}_z$
Cylindrical	$\nabla V = \frac{\partial V}{\partial \rho} \hat{a}_\rho + \frac{1}{\rho} \frac{\partial V}{\partial \phi} \hat{a}_\phi + \frac{\partial V}{\partial z} \hat{a}_z$
Spherical	$\nabla V = \frac{\partial V}{\partial r} \hat{a}_r + \frac{1}{r} \frac{\partial V}{\partial \theta} \hat{a}_\theta + \frac{1}{r \sin \theta} \frac{\partial V}{\partial \phi} \hat{a}_\phi$

Properties of the Gradient

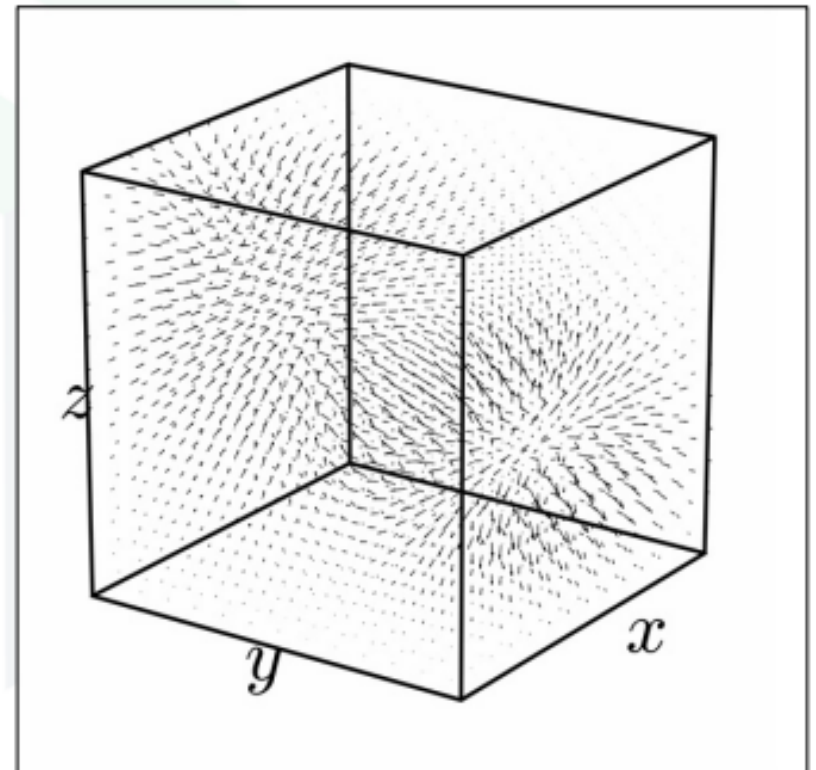
1. The gradient of a scalar function is a vector function.
2. The magnitude of ∇V is the local maximum rate of change in V .
3. ∇V points in the direction of maximum rate of change in V .
4. ∇V at any point is perpendicular to the constant V surface that passes through that point.
5. ∇V points toward increasing numbers in V .



Divergence of a Vector Field (1 of 2)

Start with the following vector field $\vec{F}(x, y, z)$.

Observe how the field seems to be converging to point in the upper left and diverging from a point in the low right.

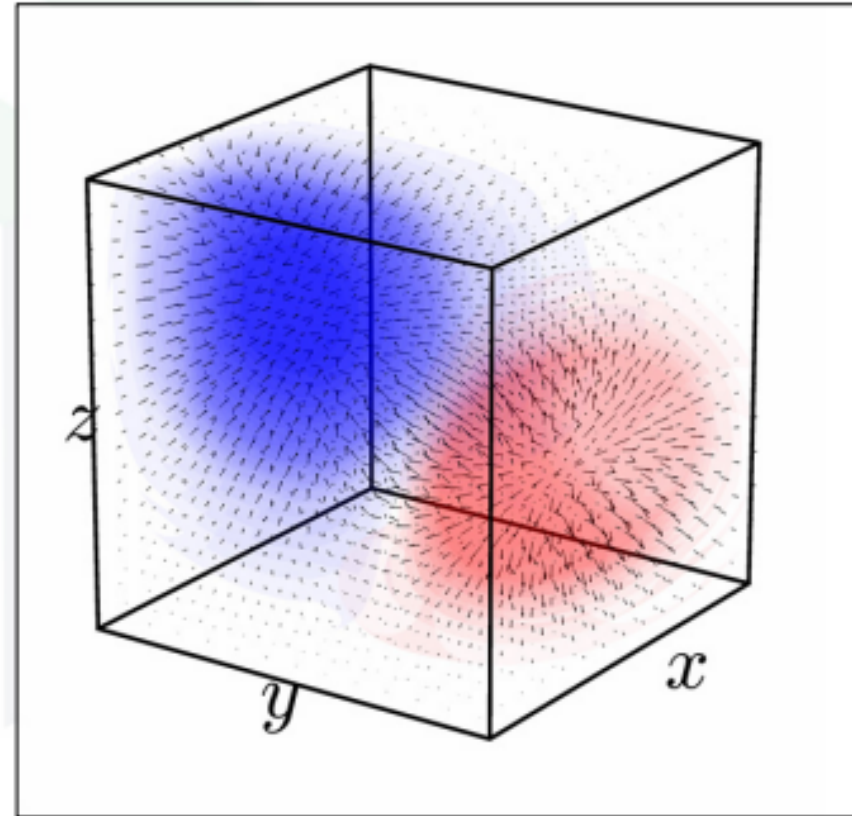


Divergence of a Vector Field (2 of 2)

The divergence is $\nabla \cdot \vec{F}(x, y, z)$.

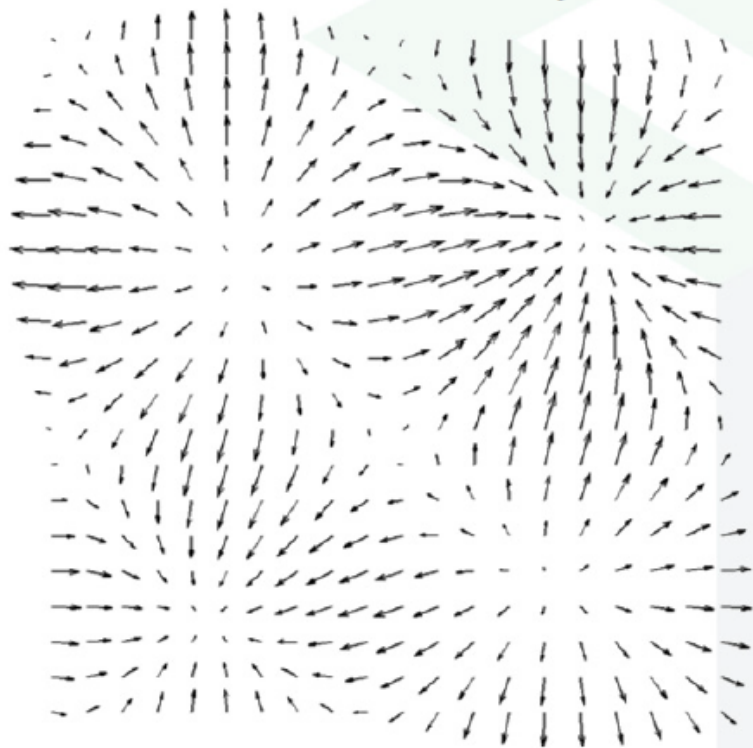
Divergence measures the tendency of a vector field to diverge from a point or converge to a point.

Divergence is a scalar quantity and is shown as the colored regions. **Blue** indicates negative divergence (convergence) and **red** indicates positive divergence.

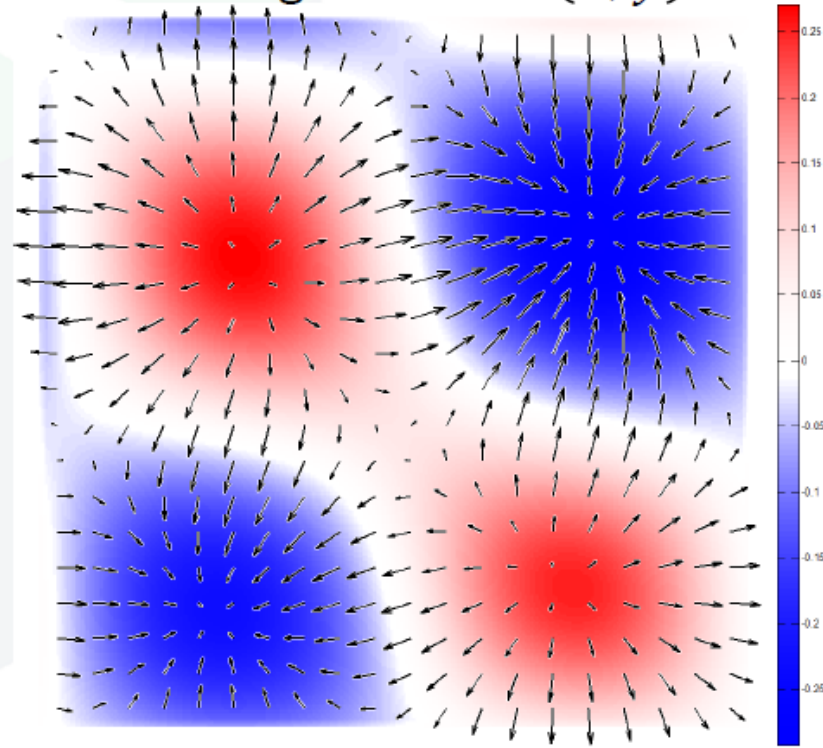


Divergence of a Vector Field (2D)

Vector Field $\vec{A}(x, y)$



Divergence $\nabla \cdot \vec{A}(x, y)$



Divergence $\nabla \cdot \vec{A}$

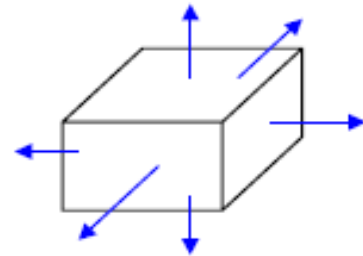
The divergence of a vector field is a scalar field that measures the tendency of a vector field to diverge from a point or converge to a point.

Coordinate System	Divergence Operation
Cartesian	$\nabla \bullet \vec{A} = \frac{\partial A_x}{\partial x} + \frac{\partial A_y}{\partial y} + \frac{\partial A_z}{\partial z}$
Cylindrical	$\nabla \bullet \vec{A} = \frac{1}{\rho} \frac{\partial (\rho A_\rho)}{\partial \rho} + \frac{1}{\rho} \frac{\partial A_\phi}{\partial \phi} + \frac{\partial A_z}{\partial z}$
Spherical	$\nabla \bullet \vec{A} = \frac{1}{r^2} \frac{\partial (r^2 A_r)}{\partial r} + \frac{1}{r \sin \theta} \frac{\partial (A_\theta \sin \theta)}{\partial \theta} + \frac{1}{r \sin \theta} \frac{\partial A_\phi}{\partial \phi}$

Divergence of a Vector Field

Divergence of a vector field $\mathbf{A}$:

$$\text{div } \mathbf{A} \equiv \lim_{\Delta v \rightarrow 0} \frac{\oint_S \mathbf{A} \cdot d\mathbf{S}}{\Delta v}$$



If we consider the vector field $\mathbf{A}$ as a flux density (per unit surface area), the closed surface integral represents the net flux leaving the volume Δv

In rectangular coordinates,

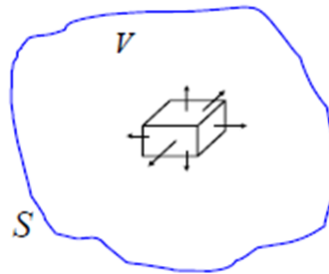
$$\text{div } \mathbf{A} = \nabla \cdot \mathbf{A} = \frac{\partial A_x}{\partial x} + \frac{\partial A_y}{\partial y} + \frac{\partial A_z}{\partial z}$$

Divergence Theorem

If $\mathbf{A}$ is a vector, then for a volume V surrounded by a closed surface S ,

$$\int_V \nabla \cdot \mathbf{A} \, dv = \oint_S \mathbf{A} \cdot d\mathbf{S}$$

The above integral represents the net flux leaving the closed surface S if $\mathbf{A}$ is the flux density



Divergence Theorem

$$\oint_S \vec{F} \cdot d\vec{S} = \int_V \vec{\nabla} \cdot \vec{F} dV$$

proof p 119

Gauss' Laws in Point Form

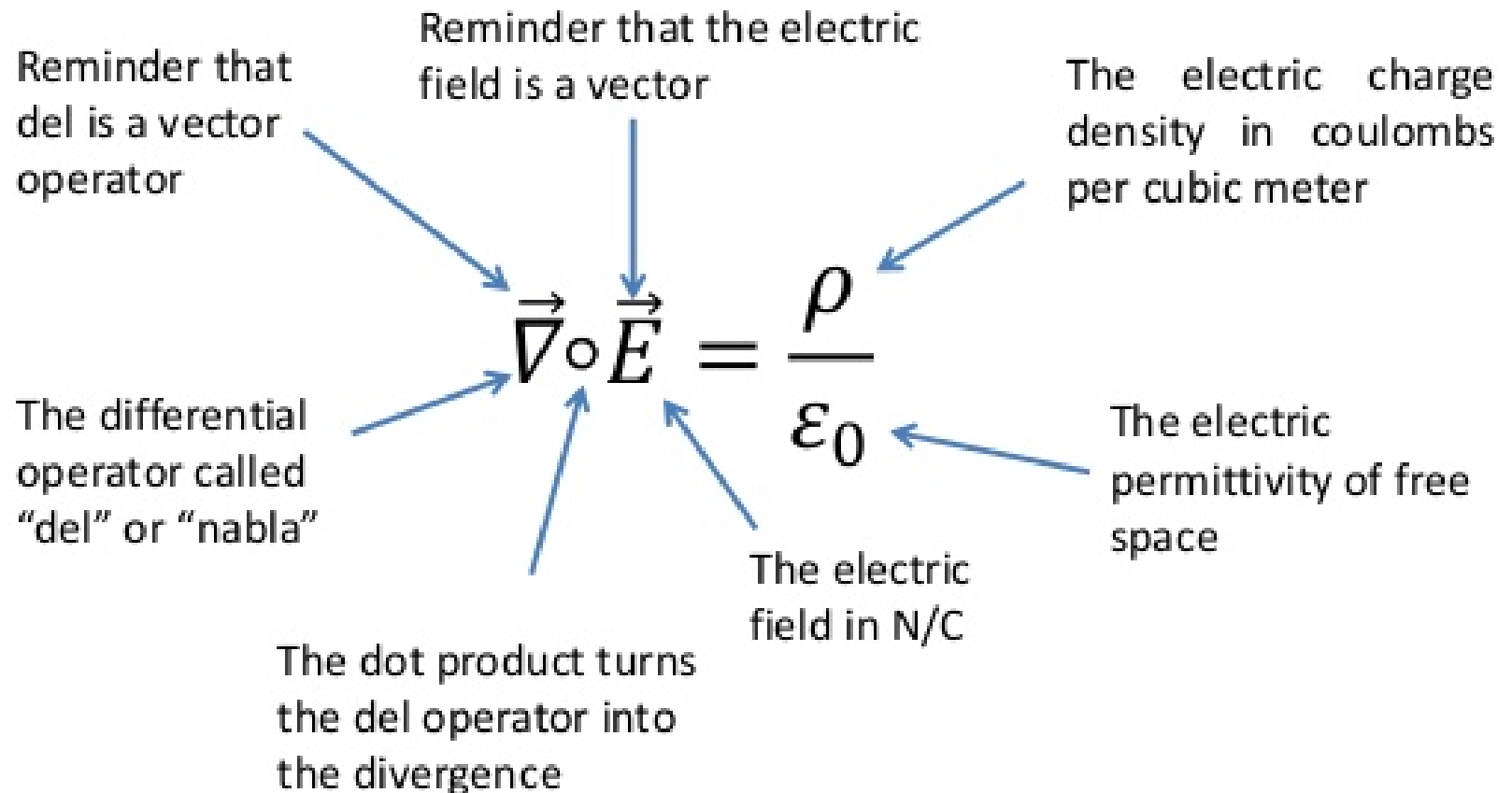
$$\left. \begin{aligned} \oint_S \epsilon_0 \vec{E} \cdot d\vec{S} &= \int_V \vec{\nabla} \cdot (\epsilon_0 \vec{E}) dV \\ \oint_S \epsilon_0 \vec{E} \cdot d\vec{S} &= \int_V \rho_v dV \end{aligned} \right\}$$

$$\Rightarrow \boxed{\vec{\nabla} \cdot (\epsilon_0 \vec{E}) = \rho_v}$$

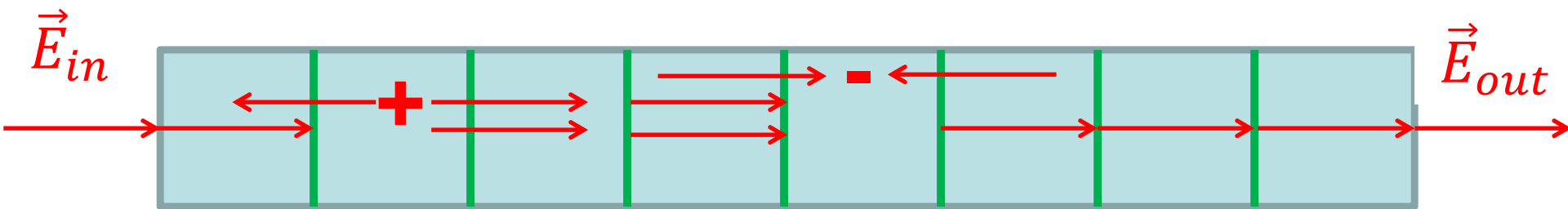
$$\boxed{\vec{\nabla} \cdot \vec{B} = 0}$$

Gauss's law for electric fields

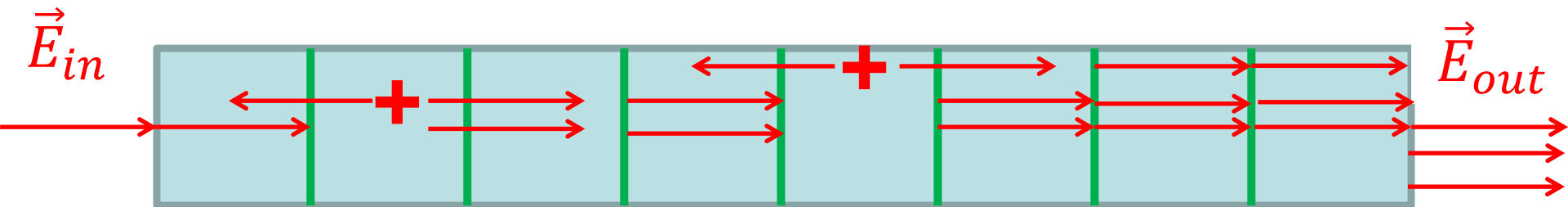
Differential form



1D Example



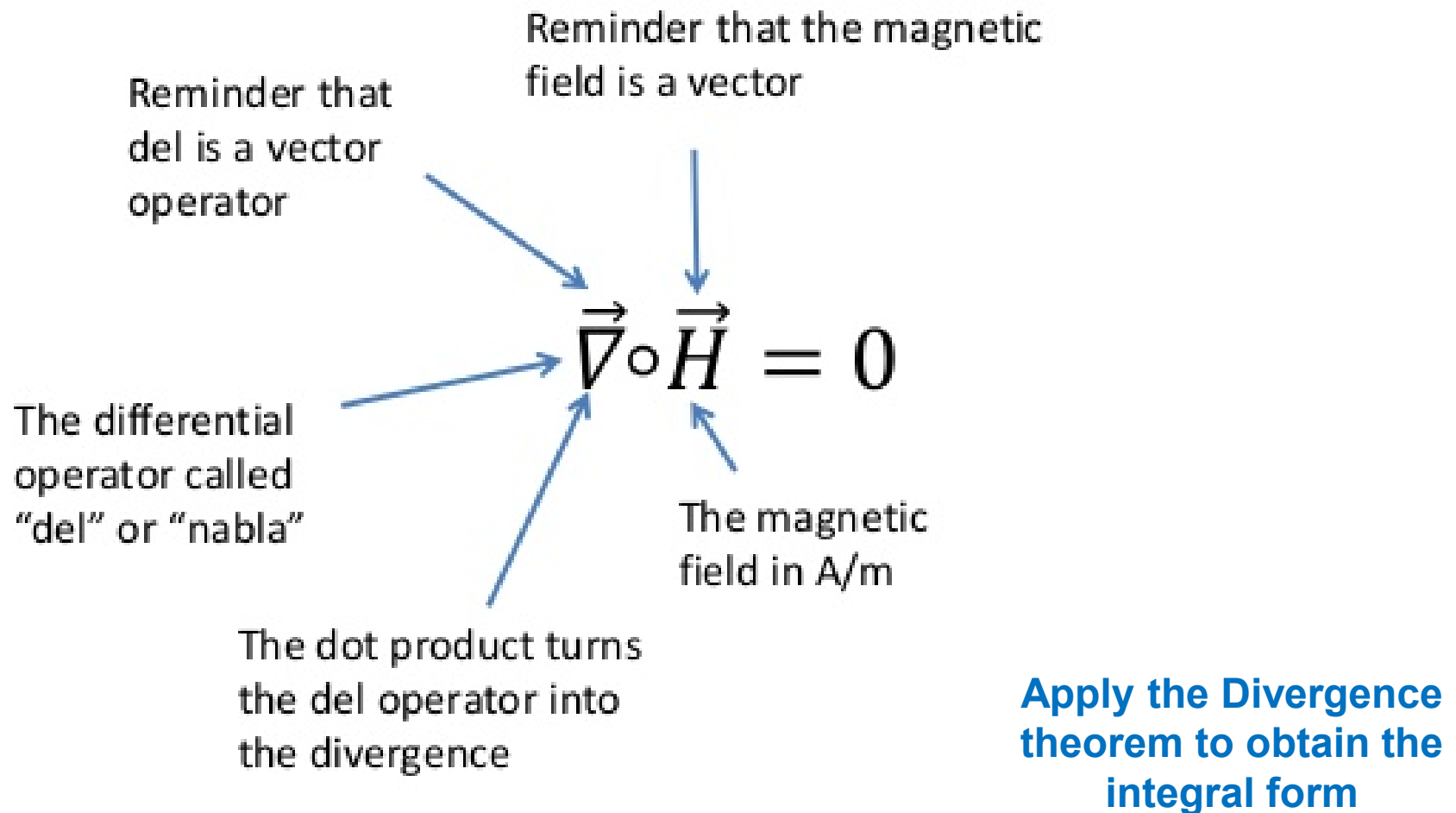
$$\vec{E}_{in} = \vec{E}_{out}$$



$$\vec{E}_{in} \neq \vec{E}_{out}$$

Gauss's law for magnetic fields

Differential form



$$\begin{aligned}\vec{\nabla} \cdot \vec{F} &= \left(\vec{a}_x \frac{\partial}{\partial x} + \vec{a}_y \frac{\partial}{\partial y} + \vec{a}_z \frac{\partial}{\partial z} \right) (F_x \vec{a}_x + F_y \vec{a}_y + F_z \vec{a}_z) \\ &= \frac{\partial F_x}{\partial x} + \frac{\partial F_y}{\partial y} + \frac{\partial F_z}{\partial z}\end{aligned}$$

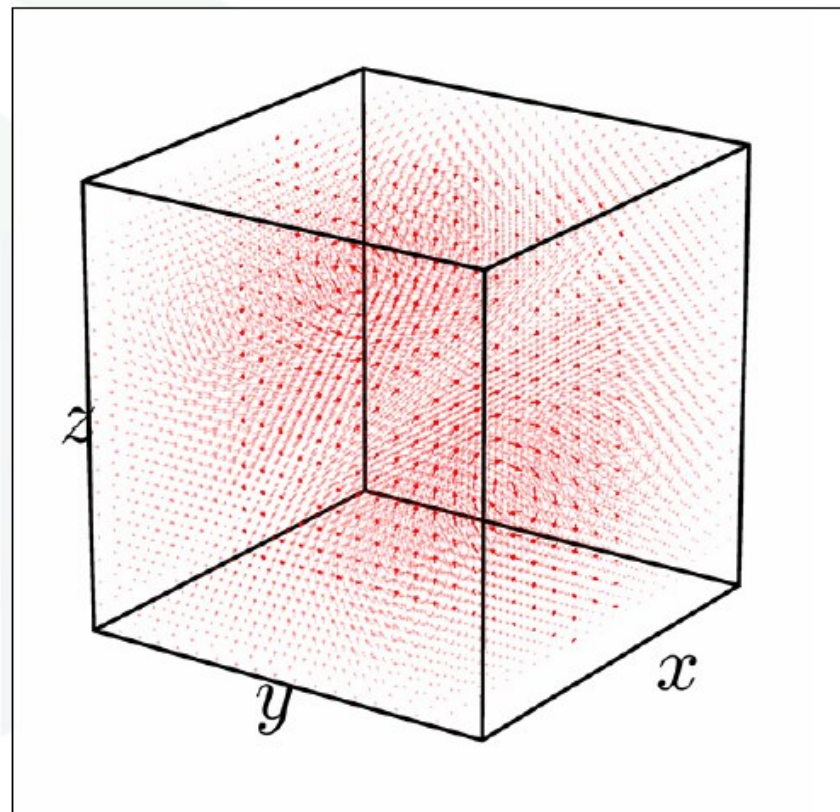
$$\text{div } \vec{F} = \frac{1}{\rho} \frac{\partial(F_\rho \rho)}{\partial \rho} + \frac{1}{\rho} \frac{\partial F_\phi}{\partial \phi} + \frac{\partial F_z}{\partial z} \quad (\text{cylind.})$$

$$\begin{aligned}\text{div } \vec{F} &= \frac{1}{r^2} \frac{\partial(F_r r^2)}{\partial r} + \frac{1}{r \sin \theta} \frac{\partial(F_\theta \sin \theta)}{\partial \theta} + \\ &\quad + \frac{1}{r \sin \theta} \frac{\partial F_\phi}{\partial \phi} \quad (\text{spherical})\end{aligned}$$

Curl of a Vector Field (1 of 2)

Start with the following vector field $\vec{F}(x, y, z)$.

Observe how the field seems to be rotating around two different axes.

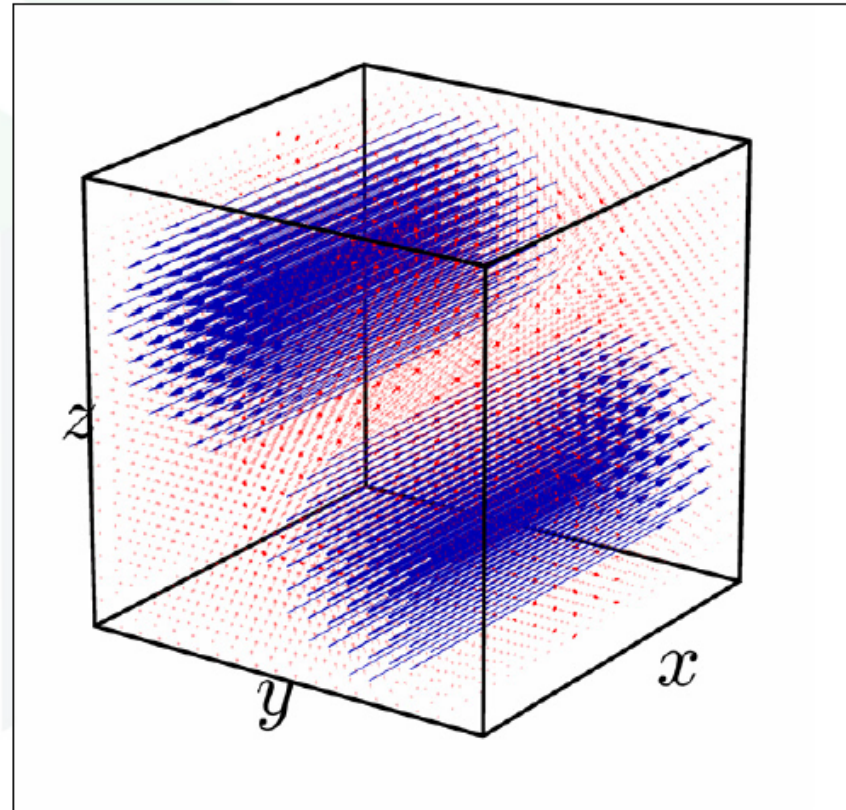


Curl of a Vector Field (1 of 2)

The curl is $\nabla \times \vec{F}(x, y, z)$.

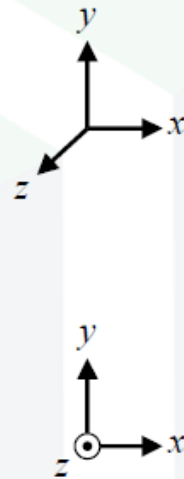
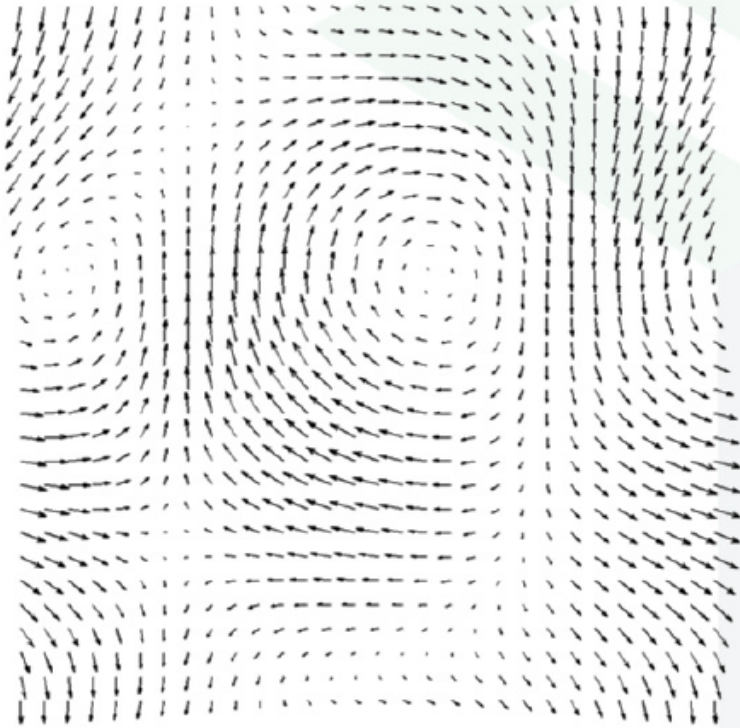
Curl measures the tendency of the vector field to circulate around an axis.

Curl is a vector quantity and is shown as the blue arrows. The magnitude of the curl conveys the strength of the circulation. The direction of the curl is the axis of the circulation.

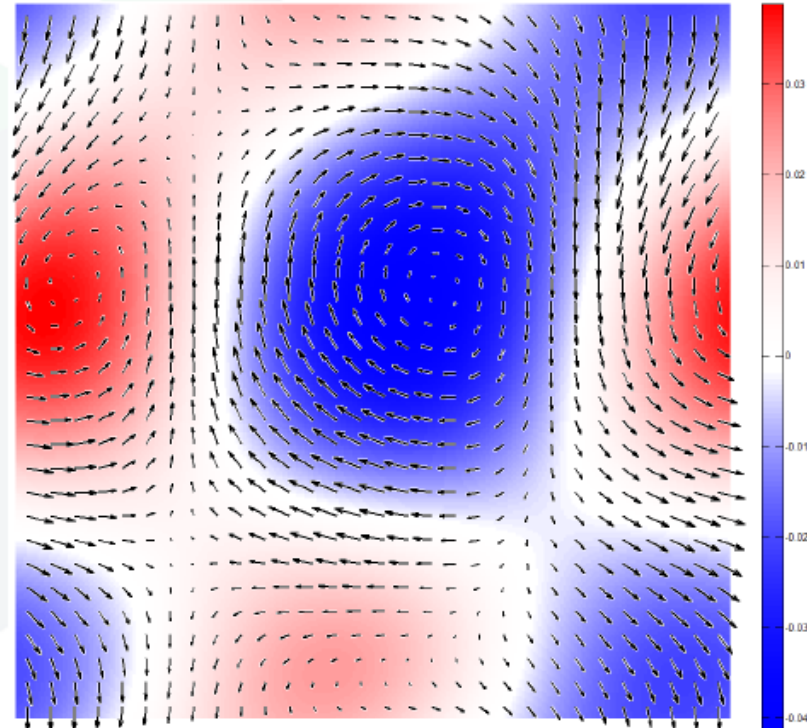


Curl of a Vector Field (~2D)

Vector Field $\vec{A}(x, y)$



Curl $\nabla \times \vec{A}(x, y)$



Curl

The curl of a vector function is a vector function that quantifies the tendency of the vector field to circulate around an axis. The magnitude of the curl conveys the strength of the circulation. The direction of the curl is the axis of the circulation.

Coordinate System	Curl Operation
Cartesian	$\nabla \times \vec{A} = \left(\frac{\partial A_z}{\partial y} - \frac{\partial A_y}{\partial z} \right) \hat{a}_x + \left(\frac{\partial A_x}{\partial z} - \frac{\partial A_z}{\partial x} \right) \hat{a}_y + \left(\frac{\partial A_y}{\partial x} - \frac{\partial A_x}{\partial y} \right) \hat{a}_z$
Cylindrical	$\nabla \times \vec{A} = \left(\frac{1}{\rho} \frac{\partial A_z}{\partial \phi} - \frac{\partial A_\phi}{\partial z} \right) \hat{a}_\rho + \left(\frac{\partial A_\rho}{\partial z} - \frac{\partial A_z}{\partial \rho} \right) \hat{a}_\phi + \frac{1}{\rho} \left(\frac{\partial (\rho A_\phi)}{\partial \rho} - \frac{\partial A_\rho}{\partial \phi} \right) \hat{a}_z$
Spherical	$\nabla \times \vec{A} = \frac{1}{r \sin \theta} \left[\frac{\partial (A_\phi \sin \theta)}{\partial \theta} - \frac{\partial A_\theta}{\partial \phi} \right] \hat{a}_r + \frac{1}{r} \left[\frac{1}{\sin \theta} \frac{\partial A_r}{\partial \phi} - \frac{\partial (r A_\phi)}{\partial r} \right] \hat{a}_\theta + \frac{1}{r} \left[\frac{\partial (r A_\theta)}{\partial r} - \frac{\partial A_r}{\partial \theta} \right] \hat{a}_\phi$

For Cartesian coordinate:

$$\nabla \times \mathbf{A} = \begin{vmatrix} \mathbf{a}_x & \mathbf{a}_y & \mathbf{a}_z \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ A_x & A_y & A_z \end{vmatrix}$$

$$\nabla \times \mathbf{A} = \left[\frac{\partial A_z}{\partial y} - \frac{\partial A_y}{\partial z} \right] \mathbf{a}_x - \left[\frac{\partial A_z}{\partial x} - \frac{\partial A_x}{\partial z} \right] \mathbf{a}_y + \left[\frac{\partial A_y}{\partial x} - \frac{\partial A_x}{\partial y} \right] \mathbf{a}_z$$

For Circular cylindrical coordinate:

$$\nabla \times \mathbf{A} = \frac{1}{\rho} \begin{vmatrix} \mathbf{a}_\rho & \rho \mathbf{a}_\phi & \mathbf{a}_z \\ \frac{\partial}{\partial \rho} & \frac{\partial}{\partial \phi} & \frac{\partial}{\partial z} \\ A_\rho & \rho A_\phi & A_z \end{vmatrix}$$

$$\begin{aligned} \nabla \times \mathbf{A} = & \left[\frac{1}{\rho} \frac{\partial A_z}{\partial \phi} - \frac{\partial A_\phi}{\partial z} \right] \mathbf{a}_\rho - \left[\frac{\partial A_z}{\partial \rho} - \frac{\partial A_\rho}{\partial z} \right] \mathbf{a}_\phi \\ & + \frac{1}{\rho} \left[\frac{\partial (\rho A_\phi)}{\partial \rho} - \frac{\partial A_\rho}{\partial \phi} \right] \mathbf{a}_z \end{aligned}$$

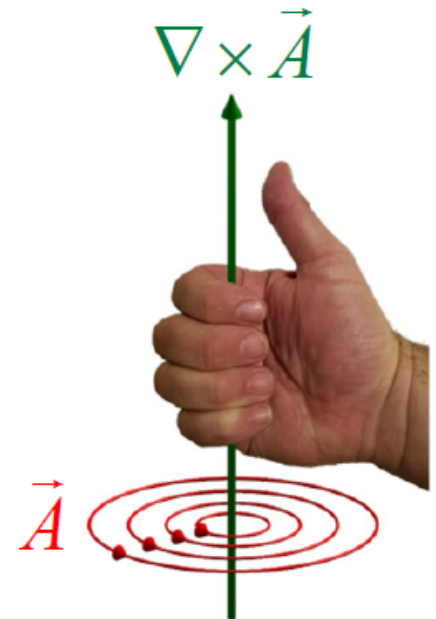
For Spherical coordinate:

$$\nabla \times \mathbf{A} = \frac{1}{r^2 \sin \theta} \begin{vmatrix} \mathbf{a}_r & \mathbf{a}_\theta & \mathbf{a}_\phi \\ \frac{\partial}{\partial r} & \frac{\partial}{\partial \theta} & \frac{\partial}{\partial \phi} \\ A_r & rA_\theta & (r \sin \theta)A_\phi \end{vmatrix}$$

$$\begin{aligned} \nabla \times \mathbf{A} = & \frac{1}{r \sin \theta} \left(\frac{\partial (\sin \theta A_\phi)}{\partial \theta} - \frac{\partial A_\theta}{\partial \phi} \right) \mathbf{a}_r + \frac{1}{r} \left(\frac{1}{\sin \theta} \frac{\partial A_r}{\partial \phi} - \frac{\partial (r A_\phi)}{\partial r} \right) \mathbf{a}_\theta \\ & + \frac{1}{r} \left(\frac{\partial (r A_\theta)}{\partial r} - \frac{\partial (A_r)}{\partial \theta} \right) \mathbf{a}_\phi \end{aligned}$$

Properties of Curl

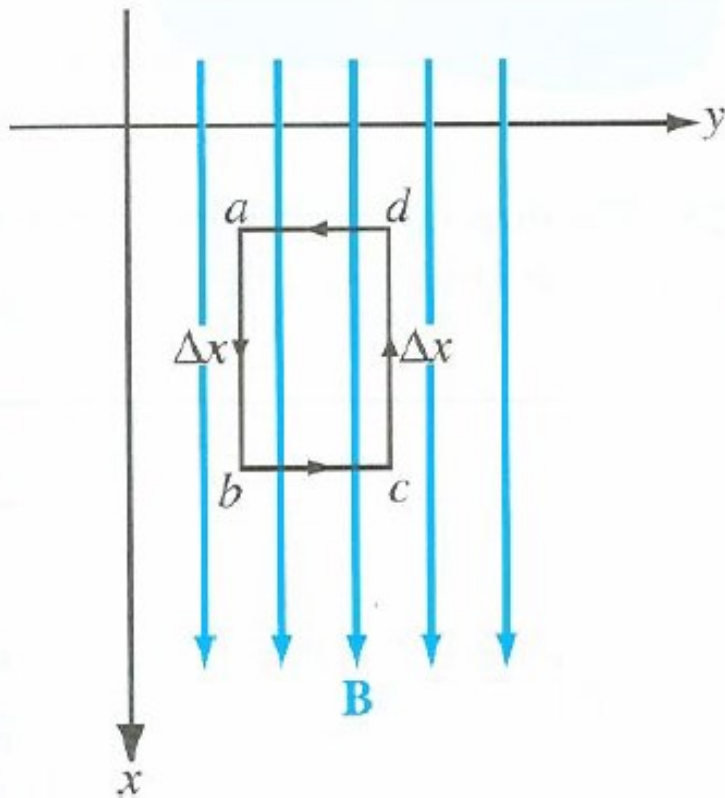
1. The curl of a vector function is a vector function.
2. The curl of a scalar field does not make sense.
3. Curl follows the right-hand rule.
4. The divergence of the curl of a vector field is always zero. $\nabla \cdot (\nabla \times \vec{A}) = 0$
5. The curl of the gradient of a scalar field is always zero. $\nabla \times (\nabla V) = 0$
6. Note that $\nabla \cdot \vec{A} \neq \vec{A} \cdot \nabla$. $\nabla \cdot \vec{A}$ calculates the derivative of $\vec{A}$, whereas $\vec{A} \cdot \nabla$ sets up a derivative operation that is scaled by $\vec{A}$.



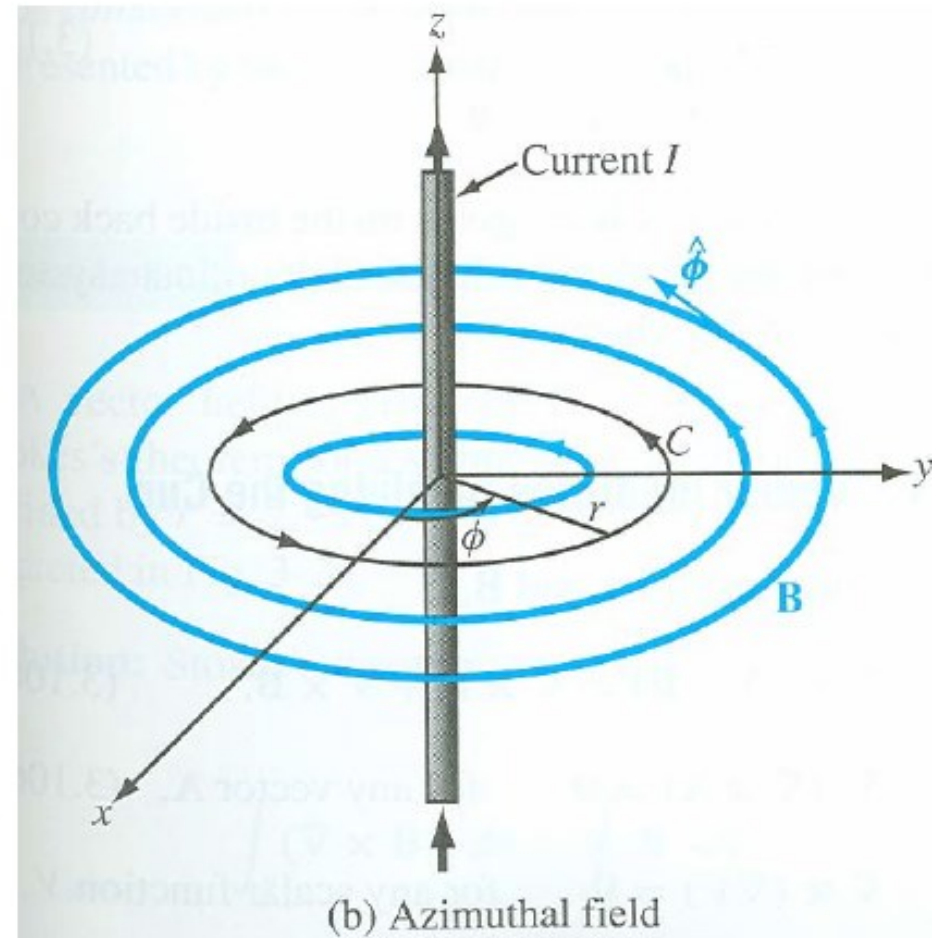
Curl of a Vector

The curl of a vector field describes the rotational property, or the circulation of the vector field.

Examples:

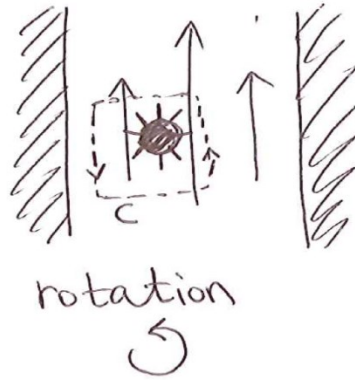
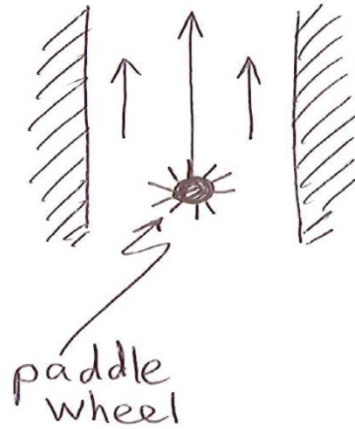


(a) Uniform field



(b) Azimuthal field

Curl of Vector Field (2)



$$\Rightarrow \oint_C \vec{F} \cdot d\vec{e} \neq 0$$

$$\text{curl } \vec{F} = [\text{curl } \vec{F}]_x \vec{a}_x + [\text{curl } \vec{F}]_y \vec{a}_y + [\text{curl } \vec{F}]_z \vec{a}_z$$

$$[\text{curl } \vec{F}]_x = \lim_{\substack{\Delta y \rightarrow 0 \\ \Delta z \rightarrow 0}} \frac{\oint_C \vec{F} \cdot d\vec{e}}{\Delta y \Delta z}$$

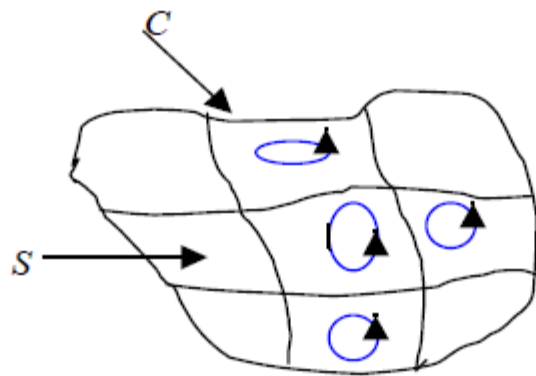
$$\text{curl } \vec{F} = \begin{vmatrix} \vec{a}_x & \vec{a}_y & \vec{a}_z \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ F_x & F_y & F_z \end{vmatrix} = \vec{\nabla} \times \vec{F}$$

"The curl of a vector is a point form expression of the line integral of a vector field around a closed contour"

Stokes's Theorem

Stokes's theorem: For an open surface S bounded by a contour C ,

$$\int_S (\nabla \times \mathbf{A}) \cdot d\mathbf{S} = \oint_C \mathbf{A} \cdot d\mathbf{l}$$



The line integrals from adjacent cells cancel leaving the only the contribution along the contour C which bounds the surface S .

Stokes' Theorem

(3)

$$\boxed{\int_S \vec{\nabla} \times \vec{F} \cdot d\vec{S} = \oint_C \vec{F} \cdot d\vec{e}}$$

proof p 136

Ampere's law in integral form:

$$\oint_C \frac{\vec{B}}{\mu_0} \cdot d\vec{e} = \int_S \vec{J} \cdot d\vec{S} + \frac{d}{dt} \int_S \epsilon_0 \vec{E} \cdot d\vec{S}$$

$$\int_S \vec{\nabla} \times \frac{\vec{B}}{\mu_0} \cdot d\vec{S} = \int_S \vec{J} \cdot d\vec{S} + \frac{d}{dt} \int_S \epsilon_0 \vec{E} \cdot d\vec{S}$$

$$\boxed{\text{curl } \frac{\vec{B}}{\mu_0} = \vec{J} + \frac{\partial \epsilon_0 \vec{E}}{\partial t}}$$

$$\boxed{\vec{\nabla} \times \frac{\vec{B}}{\mu_0} = \vec{J}}$$

static ↑

Faraday's law:

$$\boxed{\vec{\nabla} \times \vec{E} \equiv \text{curl } \vec{E} = -\frac{\partial \vec{B}}{\partial t}}$$

$$\boxed{\vec{\nabla} \times \vec{E} = 0}$$

static

Faraday's law

Differential form

The diagram illustrates the differential form of Faraday's law, $\vec{\nabla} \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$. The equation is centered, with blue arrows pointing from descriptive text to its components. The text 'Reminder that del is a vector operator' points to the $\vec{\nabla}$ symbol. The text 'Reminder that the electric field is a vector' points to the $\vec{E}$ symbol. The text 'The differential operator called "del" or "nabla"' points to the ∇ symbol. The text 'The cross-product turns the del operator into the curl' points to the $\times$ symbol. The text 'The electric field in V/m' points to the E symbol. The text 'The rate of change of the magnetic induction with time' points to the $\frac{\partial \vec{B}}{\partial t}$ term.

Reminder that del is a vector operator

Reminder that the electric field is a vector

The differential operator called "del" or "nabla"

The cross-product turns the del operator into the curl

The electric field in V/m

The rate of change of the magnetic induction with time

$$\vec{\nabla} \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$$

The Ampere-Maxwell law

Differential form

The diagram illustrates the Ampere-Maxwell law in differential form, $\vec{\nabla} \times \vec{H} = \left(\vec{J} + \epsilon_0 \frac{\partial \vec{E}}{\partial t} \right)$. Blue arrows point from descriptive text to specific parts of the equation:

- Reminder that the magnetic field is a vector**: Points to $\vec{H}$.
- Reminder that the current density is a vector**: Points to $\vec{J}$.
- The electric permittivity of free space**: Points to ϵ_0 .
- The rate of change of the electric field with time**: Points to $\frac{\partial \vec{E}}{\partial t}$.
- Reminder that the dell operator is a vector**: Points to $\vec{\nabla}$.
- The differential operator called "del" or "nabla"**: Points to $\vec{\nabla}$.
- The magnetic field in A/m**: Points to $\vec{H}$.
- The electric current density in amperes per square meter**: Points to $\vec{J}$.
- The cross-product turns the del operator into the curl**: Points to the cross-product symbol $\times$.

$$\vec{\nabla} \times \vec{H} = \left(\vec{J} + \epsilon_0 \frac{\partial \vec{E}}{\partial t} \right)$$

Maxwell's Equations

INTEGRAL FORMS

$$\oint_s \epsilon_o \epsilon_r \mathbf{E} \cdot d\mathbf{s} = \int_v \rho_v dv$$

$$\oint_s \mathbf{B} \cdot d\mathbf{s} = 0$$

$$\oint_c \mathbf{E} \cdot d\boldsymbol{\ell} = -\frac{d}{dt} \int_s \mathbf{B} \cdot d\mathbf{s}$$

$$\oint_c \mathbf{H} \cdot d\boldsymbol{\ell} = \int_s \mathbf{J} \cdot d\mathbf{s} + \frac{d}{dt} \int_s \epsilon_o \epsilon_r \mathbf{E} \cdot d\mathbf{s}$$

DIFFERENTIAL FORMS

$$\boldsymbol{\nabla} \cdot \epsilon_o \epsilon_r \mathbf{E} = \rho_v \quad (4.1)$$

$$\boldsymbol{\nabla} \cdot \mathbf{B} = 0 \quad (4.2)$$

$$\boldsymbol{\nabla} \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t} \quad (4.3)$$

$$\boldsymbol{\nabla} \times \mathbf{H} = \mathbf{J} + \frac{\partial \epsilon_o \epsilon_r \mathbf{E}}{\partial t} \quad (4.4)$$

(4)

Properties of a gradient

- The grad of a scalar field is a vector perpendicular to the surface characterized by a constant value of the scalar field.
- Magnitude of grad equal to the maximum rate of change of the scalar field.
- The line integral of the gradient of any scalar function Φ over any closed path is zero:

$$\oint \vec{\nabla} \Phi \cdot d\vec{r} = 0 \quad (\text{conservative property})$$

$$\Rightarrow \boxed{\vec{E} = -\nabla \Phi}$$

Φ - electrostatic potential

$$\Phi_1 = \int_a^0 \vec{E} \cdot d\vec{e}$$

Electrostatic Potential

(5)

$$\vec{E} = -\left(\vec{a}_x \frac{\partial \Phi}{\partial x} + \vec{a}_y \frac{\partial \Phi}{\partial y} + \vec{a}_z \frac{\partial \Phi}{\partial z}\right) \quad (\text{c.c.s.})$$

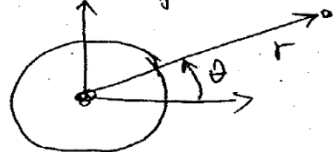
$$\vec{E} \equiv -\vec{\nabla} \Phi = -\left[\vec{a}_\rho \frac{\partial \Phi}{\partial \rho} + \vec{a}_\phi \frac{1}{\rho} \frac{\partial \Phi}{\partial \phi} + \vec{a}_z \frac{\partial \Phi}{\partial z}\right]$$

(cylind. c.s.)

$$\vec{E} \equiv -\vec{\nabla} \Phi = -\left[\vec{a}_r \frac{\partial \Phi}{\partial r} + \vec{a}_\theta \frac{1}{r} \frac{\partial \Phi}{\partial \theta} + \vec{a}_\phi \frac{1}{r \sin \theta} \frac{\partial \Phi}{\partial \phi}\right]$$

(spherical c.s.)

Example



$$\Phi \equiv \Phi(r, \theta) = E_0 \left[1 - \left(\frac{a}{r}\right)^3\right] r \cos \theta$$

$\vec{E} = ?$

$$\vec{E} \equiv -\vec{\nabla} \Phi = -\vec{a}_r \frac{\partial \Phi}{\partial r} - \vec{a}_\theta \frac{1}{r} \frac{\partial \Phi}{\partial \theta} =$$

$$= -\vec{a}_r E_0 \left[1 + 2\left(\frac{a}{r}\right)^3\right] \cos \theta + \vec{a}_\theta E_0 \left[1 - \left(\frac{a}{r}\right)^3\right] \sin \theta$$

Conservative Property of Electrostatic Field

$$\oint_C \vec{E} \cdot d\vec{r} = 0$$

← conservative field
"the line integral along any closed contour is zero"

$$\vec{E} = -\nabla \Phi$$

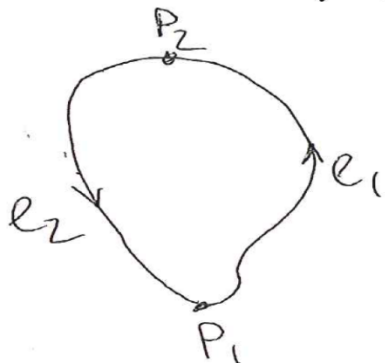
or $\vec{E} = -\text{grad } \Phi$

$$\oint_C \vec{\nabla} \Phi \cdot d\vec{r} = 0 \Leftrightarrow \oint_C d\Phi = 0$$

$$d\Phi = \frac{\partial \Phi}{\partial x} dx + \frac{\partial \Phi}{\partial y} dy + \frac{\partial \Phi}{\partial z} dz =$$

$$= \left(\frac{\partial \Phi}{\partial x} \vec{a}_x + \frac{\partial \Phi}{\partial y} \vec{a}_y + \frac{\partial \Phi}{\partial z} \vec{a}_z \right) (dx \vec{a}_x + dy \vec{a}_y + dz \vec{a}_z)$$

$$= \vec{\nabla} \Phi \cdot d\vec{r}$$



$$\oint_C d\Phi = \int_{C_1} d\Phi + \int_{C_2} d\Phi =$$

$$= \Phi \Big|_{P_1}^{P_2} + \Phi \Big|_{P_2}^{P_1} = (\Phi_{P_2} - \Phi_{P_1}) + (\Phi_{P_1} - \Phi_{P_2}) = 0$$

Work Done by Electric Field

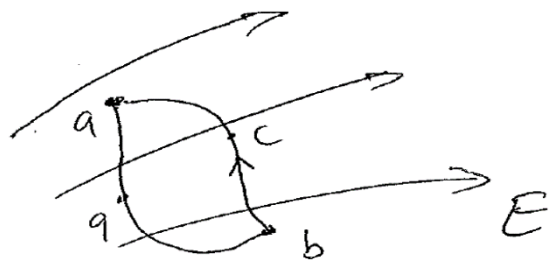
$$dW_e = \vec{F} \cdot d\vec{r} = q \vec{E} \cdot d\vec{r}$$

Work of external field: $dW = -dW_e = -q \vec{E} \cdot d\vec{r}$

Work in moving charge:

$$W_{ab} = -q \int_a^b \vec{E} \cdot d\vec{r}$$

← total work
done by external
forces



Lines of
force

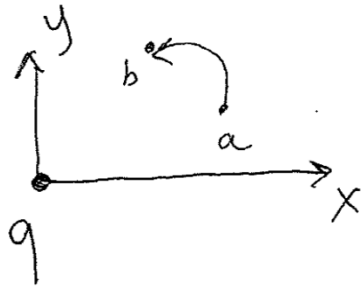
$$W_{ab} = +q \int_{\Phi_b}^{\Phi_a} d\Phi =$$

$$= q(\Phi_a - \Phi_b) =$$

$$= qV_{ab}$$

Does not depend on
a particular path

Point charge



$$\vec{E} = \frac{q}{4\pi\epsilon_0 r^2} \vec{a}_r$$

$$V_{ab} = - \int_{r_2}^{r_1} \frac{q}{4\pi\epsilon_0 r^2} dr = \frac{q}{4\pi\epsilon_0} \left[\frac{1}{r_1} - \frac{1}{r_2} \right]$$

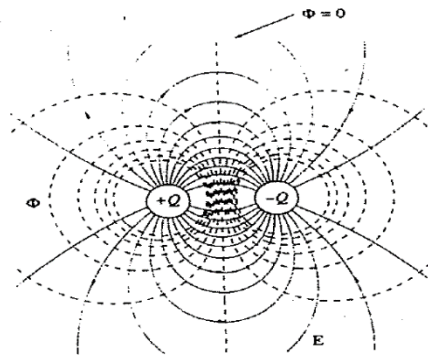
$$\text{If } r_2 \rightarrow \infty$$

$$V_a = \frac{q}{4\pi\epsilon_0 R}$$

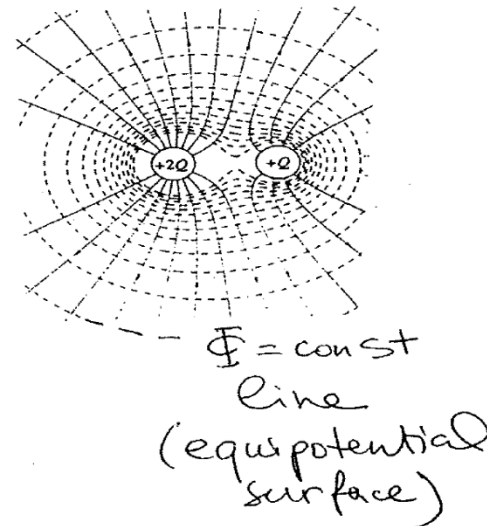
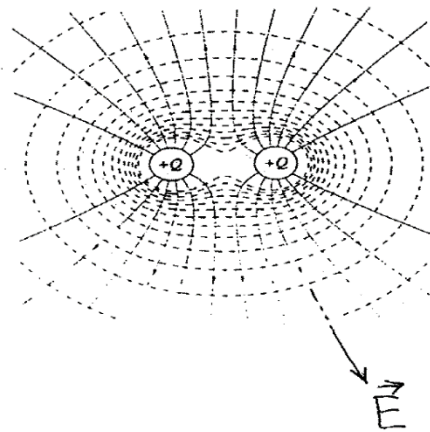
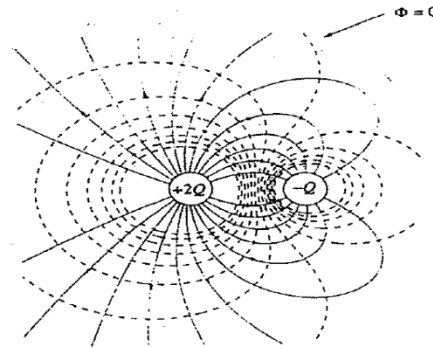
(6)

Illustration of the concept of electrostatic potential

$$\vec{E} = -\text{grad } \Phi$$



(a)



Example

(3)

Divergence - Free (Zero-Divergence) Fields

$$\vec{A}_1 = -x \vec{a}_x + y \vec{a}_y$$

$$\vec{\nabla} \cdot \vec{A}_1 = \frac{\partial}{\partial x}(-x) + \frac{\partial}{\partial y}(y) = -1 + 1 = 0$$

$$\vec{A}_2 = \frac{1}{2} x^2 \vec{a}_x - xy \vec{a}_y$$

$$\vec{\nabla} \cdot \vec{A}_2 = \frac{\partial}{\partial x} \left(\frac{1}{2} x^2 \right) + \frac{\partial}{\partial y} (-xy) = x - x = 0$$

$$\vec{A}_3 = \sin y \vec{a}_x - \cos x \vec{a}_y$$

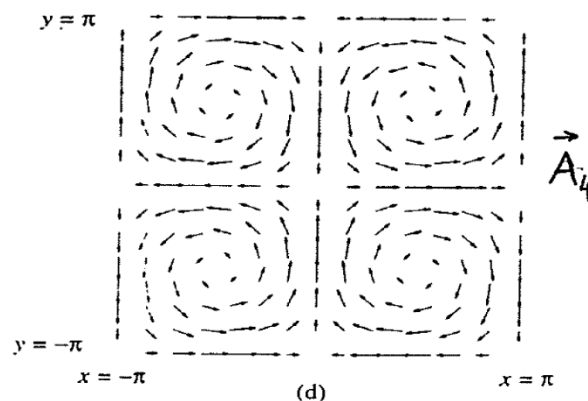
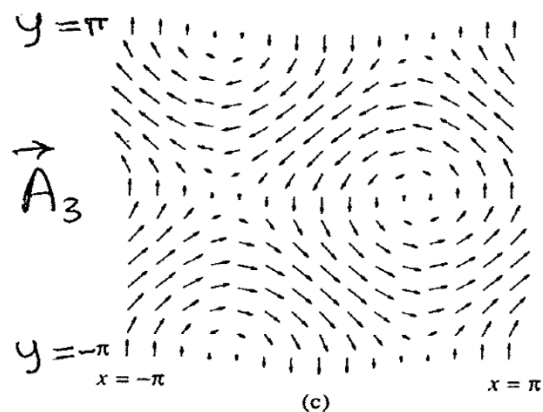
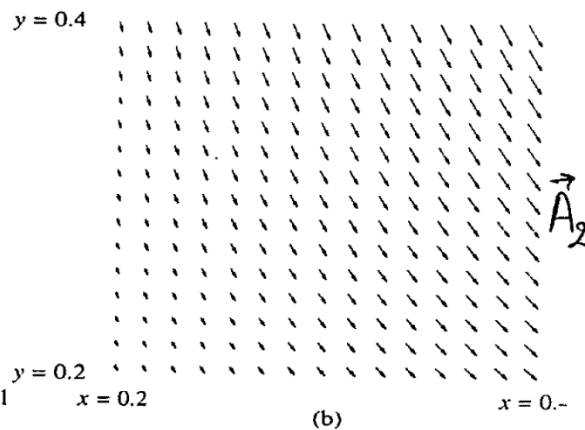
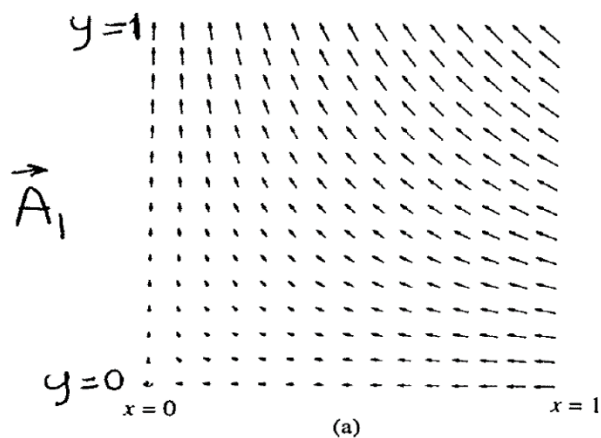
$$\vec{\nabla} \cdot \vec{A}_3 = 0 - 0 = 0$$

$$\vec{A}_4 = \sin x \cos y \vec{a}_x - \cos x \sin y \vec{a}_y$$

$$\vec{\nabla} \cdot \vec{A}_4 = \cos y \cdot \cos x - \cos x \cdot \cos y = 0$$

(4)

Divergence - Free Fields



(5)

Vector Fields with Non-Zero Divergence

$$\vec{A}_5 = \frac{x^2}{2} \vec{a}_x + xy \vec{a}_y$$

$$\vec{\nabla} \cdot \vec{A}_5 = x + x = 2x$$

$$\vec{A}_6 = \sin x \vec{a}_x - \cos y \vec{a}_y$$

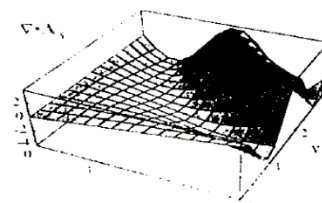
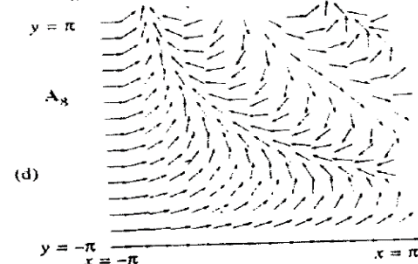
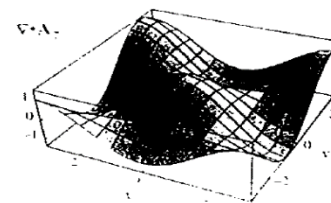
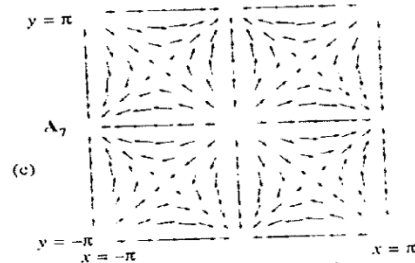
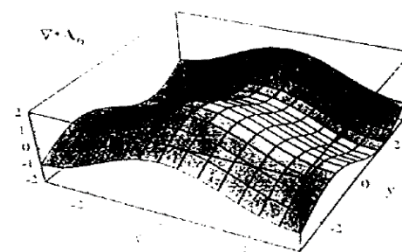
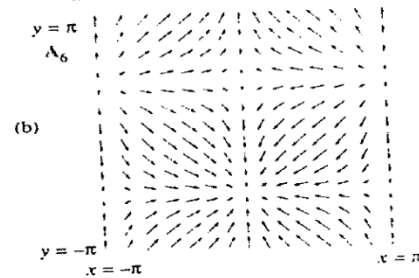
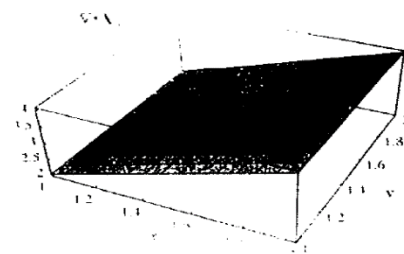
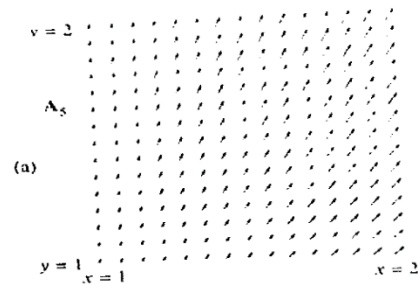
$$\vec{\nabla} \cdot \vec{A}_6 = \cos x + \sin y$$

$$\vec{A}_7 = \sin x \cos y \vec{a}_x + \cos x \sin y \vec{a}_y$$

$$\vec{A}_8 = \cos(xy) \vec{a}_x + \sin(xy) \vec{a}_y$$

Vector Fields with Non-zero Divergence

⑥



$\vec{\nabla} \cdot \epsilon_0 \vec{E} = \rho_v$